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DD FORM 1426 MS 672-1 (Continued)

P.A. NAVY — AS Other Coast AF - 85 ARMY - ER

PROCUREMENT SPECIFICATION MIL-C-5809

International interest *

SUPERSEDING

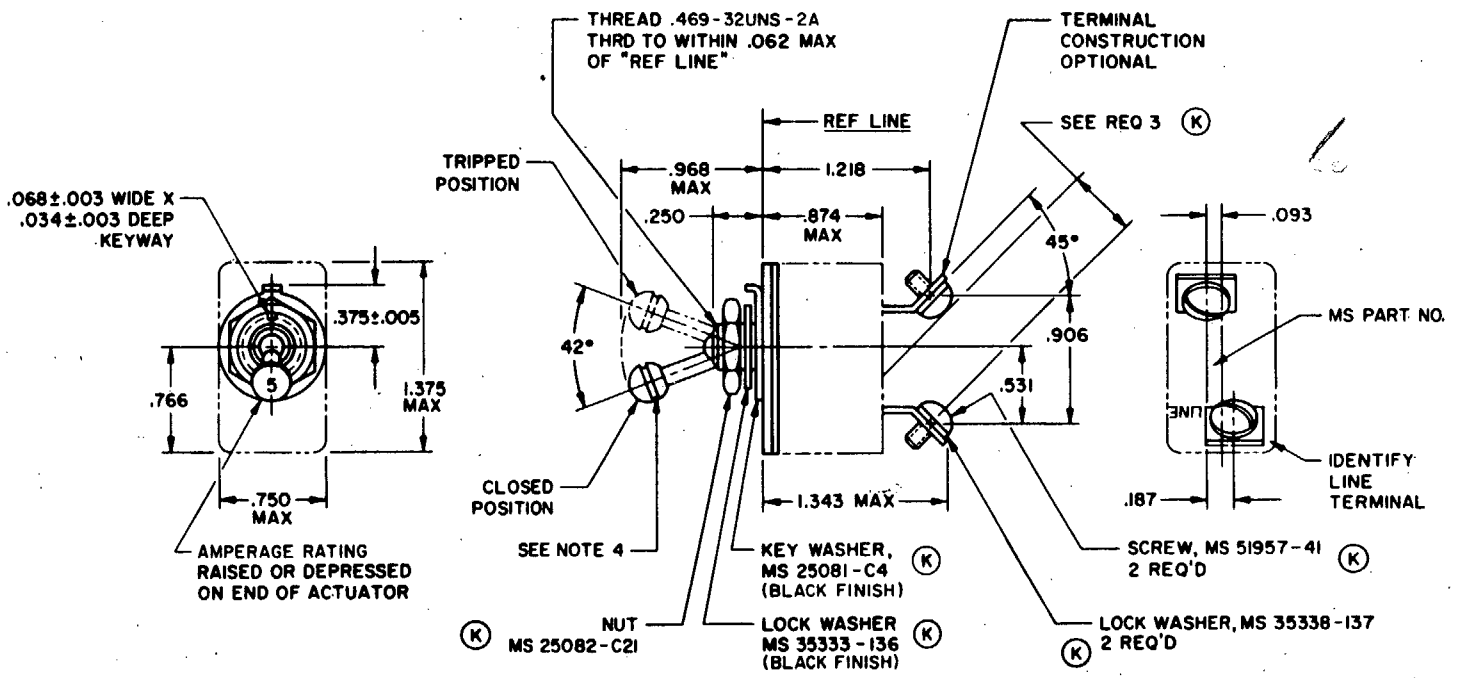
TITLE (K) Circuit Breaker - Aircraft, Trip-Free, Toggle, 5 Thru 15 Amp, Type I

MILITARY STANDARD

MS 24509

SHEET 1 OF 5

(K) DENOTES CHANGE



CONFIGURATION A (BUSHING MOUNT)

FED. SUP CLASS 5925

APPROVED 22 APR 59 REVISED (H) 12 APR 84 (J) 12 JUN 86 (K) 30 NOV 1987

PROJECT NO. 5925-0193

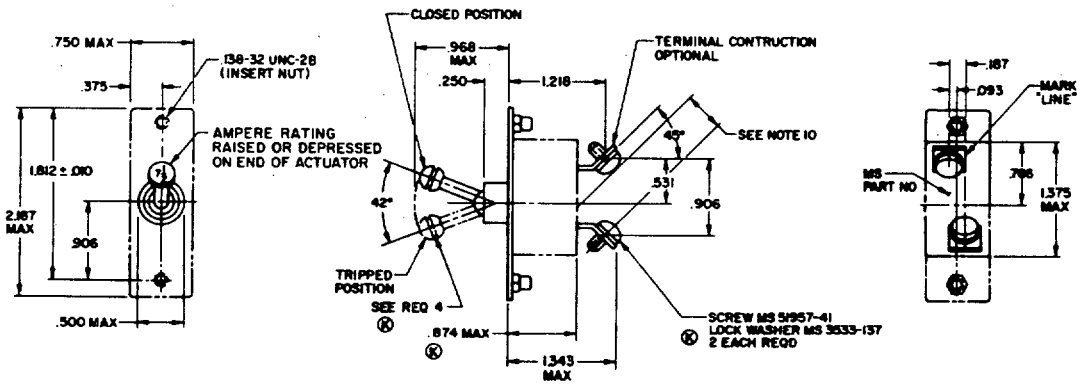
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CONFIGURATION B (COVER PLATE MOUNT)

(K) REQUIREMENTS:

1. The part numbers for circuit breakers in accordance with this specification shall conform to the example below.

MS24509 - A 5
- 5 — Amperage Rating
- A — Mounting Configuration ("A" or "B")
- MS24509 — Basic Part Style

2. Circuit breakets are recycling as defined in MIL-C-5809.
3. .425 min. typ. clearance from centerline of screw to case.
4. Groove indicates trip free action.
5. Mounting nut torque: 65 inch-pounds maximum.

(K) FOR CHANGES SEE PAGES 1, 2, 3, 4, & 5

APPROVED 22 APR 59
REVISED

P.A. NAVY — AS Other Cost: AF - 85, ARMY - ER	International interest: *	TITLE: (K) Circuit Breaker - Aircraft, Trip-Free, Toggle, 5 Thru 15 Amp, Type I	MILITARY STANDARD MS 24509
PROCUREMENT SPECIFICATION: MIL-C-5809		SUPERSEDES:	SHEET 2 OF 5

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TABLE I

Electrical and mechanical characteristics

Dash Number	Nominal Capacity Amperes	Voltage Drop Max	Weight Max Lbs	Operating force max lbs		Endurance cycles				Mech No Load
						Resistive		Inductive		
				Pullout	Reset	AC	DC	AC	DC	
-5	5	.65	.087	6	6	1/ 5,000	2/ 5,000	1/ 5,000	2/ 1,000	10,000
-7-1/2	7.5	.50	.087	6	6	5,000	5,000	.6 to	1,000	10,000
-10	10	.50	.087	6	6	5,000	5,000	.7 LAG	1,000	10,000
-15	15	.50	.087	6	6	5,000	5,000	P.F.	1,000	10,000

1/ 400 cycle 115/200 volt system, tested at 120 ± 5 volts 380-420 Hz.
2/ 28 volt dc system, tested at 30 ± 2 volts.

TABLE II

Detail calibration requirements - trip time in seconds

Dash Number	+25°C Percent rated current					-40°C Percent rated current		+71° Percent rated current	
	115	145	200	400	600	138	175	80	125
-5	NO TRIP 1 HOUR MAX	TRIP WITHIN 1 HOUR	40-100	3-22	1-10	NO TRIP 1 HOUR MAX	TRIP WITHIN 1 HOUR	NO TRIP 1 HOUP MAX	TRIP WITHIN 1 HOUR
-7-1/2			10 to 70	0.75 to 7.0	0.25 to 2.5				
-10									
-15									

Ambient temperature tolerance ±2°C.

APPROVED 22 APR 59 REVISED (K) FOR CHANGES SEE PAGES 1, 2, 3, 4, & 5

P.A. NAVY — AS Other Cust AF - 85 ARMY - ER	International interest *	TITLE (K) Circuit Breaker - Aircraft, Trip-Free, Toggle, 5 Thru 15 Amp, Type I	MILITARY STANDARD MS 24509
PROCUREMENT SPECIFICATION MIL-C-5809		SUPERSEDES:	SHEET 3 OF 5

DD FORM 1 MAR 72 672-1 (Continued) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE. PLATE NO. 22000

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TABLE III

Interruption current (amperes) requirements										
Dash number	Test designation per MIL-C-5809									
	A	B	C	D	E	F	G		H	
							1Ø	3Ø	1Ø	3Ø
-5	3,500	3,500	6,000	6,000	1,000	1,500	(*)	(*)	(*)	(*)
-7-1/2	3,500	3,500	6,000	6,000	1,000	1,500	(*)	(*)	(*)	(*)
-10	3,500	3,500	6,000	6,000	1,000	1,500	(*)	(*)	(*)	(*)
-15	3,500	3,500	6,000	6,000	1,000	1,500	(*)	(*)	(*)	(*)

(*) Not applicable

TABLE IV

ENVIRONMENTAL PERFORMANCE	
Max operating altitude - FT	70,000
Operating ambient temp range	-40°C TO +71°C
Vibration	MIL-STD-202, Method 204, Condition A
Shock	30G, MIL-STD-202, Method 213, TEST CONDITION J
Acceleration	10G

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CROSS REFERENCE For applicable cross reference see table V. The existing stocks of superseded items may be used. When exhausted, the superseding parts shall be used.

TABLE V. Cross reference.

Superseding number	Superseded numbers					
	FSCM 96906	FSCM 82647	FSCM 82647	FSCM 82647	FSCM 82647	FSCM 81349
MS24509-A-3	MS24509-3	D7270-1-3		7270-1-3		
MS24509-A-5	MS24509-5	D7270-1-5		7270-1-5		
MS24509-A-7-1/2	MS24509-7-1/2	D7270-1-7		7270-1-7		
MS24509-A-10	MS24509-10	D7270-1-10		7270-1-10		MS24509-10
MS24509-A-15	MS24509-15	D7270-1-15		7270-1-15		
MS24509-A-20	MS24509-20	D7270-1-20		7270-1-20		
MS24509-A-25	MS24509-25	D7270-1-25		7270-1-25		
MS24509-A-30	MS24509-30	D7270-1-30		7270-1-30		
MS24509-A-35	MS24509-35	D7270-1-35		7270-1-35		
MS24509-B-3		D7270-7-3			7270-7-3	
MS24509-B-5		D7270-7-5	D7270-5-5	7270-5-5	7270-7-5	
MS24509-B-7-1/2		D7270-7-7	D7270-5-7	7270-5-7	7270-7-7	
MS24509-B-10		D7270-7-10	D7270-5-10	7270-5-10	7270-7-10	
MS24509-B-15		D7270-7-15	D7270-5-10	7270-5-15	7270-7-15	
MS24509-B-20		D7270-7-20	D7270-5-20	7270-5-20	7270-7-20	
MS24509-B-25		D7270-7-25	D7270-5-25	7270-5-25	7270-7-25	
MS24509-B-30		D7270-7-30			7270-7-30	
MS24509-B-35		D7270-7-35			7270-7-35	

(K) NOTES:

1. Dimensions are in inches unless otherwise specified. Tolerance: Decimals ±.031; angles ±2°.
2. Phantom lines indicate that contour is optional provided maximum dimensions are not exceeded.
3. In the event of a conflict between the text of this standard and the references cited herein, the text of the standard shall take precedence.
4. Voltage rating: 28VDC nominal or 115V 400 Hz nominal.

*CERTAIN PROVISIONS OF THIS STANDARD ARE THE SUBJECT OF INTERNATIONAL STANDARDIZATION AGREEMENT ASCC AIR STD 12/3. WHEN REVISION OR CANCELLATION OF THIS STANDARD IS PROPOSED WHICH WILL AFFECT OR VIOLATE THE INTERNATIONAL AGREEMENT CONCERNED, THE PREPARING ACTIVITY WILL TAKE APPROPRIATE RECONCILIATION ACTION THROUGH INTERNATIONAL STANDARDIZATION CHANNELS, INCLUDING DEPARTMENTAL STANDARDIZATION OFFICES, IF REQUIRED.

5. MS24509-A-3, -20, -25, -30, -35 and MS24509-B-3, -20, -25, -30, -35 are cancelled with no superseding standard.

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